

GALENA, AK, USA

WMO: 702220

Lat: 64.737N

Lon: 156.934W

Elev: 47

StdP: 100.77

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26501

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.4	-40.7	-42.6	0.1	-37.8	-39.6	0.1	-35.0	10.5	-8.8	9.1	-8.5	0.4	90	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	25.9	16.0	23.8	15.1	22.2	14.5	17.2	23.1	16.2	21.9	15.3	20.5	2.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	10.6	18.7	13.8	9.9	17.9	12.9	9.3	17.1	48.4	23.0	45.3	21.9	42.8	20.5	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.5	6.4	DB	-44.9	28.6	2.6	1.8	-46.8	29.9	-48.4	30.9	-49.9	31.9	-51.8	33.2	(4)
			WB	-38.5	18.8	2.6	1.4	-40.4	19.7	-41.9	20.5	-43.4	21.3	-45.2	22.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.6	-21.9	-15.8	-13.0	-2.2	8.2	14.6	15.8	12.6	6.9	-2.7	-15.0	-19.8
	DBStd	15.17	11.73	9.06	7.72	6.29	4.51	3.24	2.82	3.19	4.01	6.09	8.66	9.37
	HDD10.0	5063	988	722	712	366	86	4	1	11	105	395	749	924
	HDD18.3	7665	1247	955	971	616	314	119	87	180	342	653	999	1182
	CDD10.0	454	0	0	0	0	29	141	181	91	13	0	0	0
	CDD18.3	15	0	0	0	0	0	6	8	1	0	0	0	0
	CDH23.3	210	0	0	0	0	6	79	100	26	0	0	0	0
	CDH26.7	27	0	0	0	0	1	11	13	3	0	0	0	0
Wind	WSAvg	2.3	1.8	2.2	2.4	2.7	2.6	2.4	2.6	2.5	2.5	2.5	2.1	1.8
Precipitation	PrecAvg	326	18	18	14	9	12	36	50	61	41	25	20	23
	PrecMax	462	45	42	38	29	24	70	106	102	97	54	59	61
	PrecMin	244	1	0	1	0	0	5	12	21	13	1	0	4
	PrecStd	58	11	12	10	9	7	21	21	20	22	15	15	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.8	3.0	4.6	14.4	24.1	28.0	28.6	26.5	19.9	10.9	3.3	1.7
		MCWB	1.3	1.4	1.2	6.9	12.9	16.5	17.5	17.5	12.6	7.5	1.7	0.3
	2%	DB	0.3	1.5	2.4	11.3	21.2	25.6	26.1	22.9	17.2	7.7	1.7	-0.6
		MCWB	-0.9	-0.1	0.0	5.3	11.6	15.2	16.4	15.6	10.9	5.2	0.5	-1.7
	5%	DB	-2.8	-0.6	0.6	9.0	18.9	23.6	24.0	20.7	15.2	6.2	-0.1	-3.8
		MCWB	-3.7	-1.9	-1.5	4.0	10.4	14.4	15.8	14.8	10.3	4.5	-1.1	-4.7
	10%	DB	-6.0	-3.1	-1.7	7.1	16.6	21.9	22.3	18.4	13.3	4.6	-3.0	-7.3
		MCWB	-6.6	-4.3	-3.5	3.1	9.4	13.7	15.1	13.8	9.6	3.2	-3.7	-7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	1.9	2.1	7.4	13.7	17.9	18.6	18.6	13.0	8.0	2.0	0.5
		MCDB	2.9	2.8	3.8	13.7	22.1	24.9	26.2	24.4	17.8	9.5	2.8	1.6
	2%	WB	-0.5	0.0	0.3	5.7	12.4	16.5	17.5	16.7	11.8	5.7	0.6	-1.4
		MCDB	0.6	1.3	2.2	10.5	19.7	23.1	23.6	21.3	15.8	7.1	1.5	-0.3
	5%	WB	-3.0	-1.9	-1.6	4.5	11.1	15.4	16.5	15.5	10.8	4.6	-1.2	-4.4
		MCDB	-2.1	-0.5	0.3	8.6	17.4	21.7	22.5	19.5	14.3	6.1	-0.1	-3.4
	10%	WB	-5.8	-4.2	-3.4	3.2	10.1	14.4	15.6	14.4	10.0	3.2	-3.6	-7.4
		MCDB	-5.2	-3.0	-1.6	6.8	15.5	20.3	21.1	17.6	13.0	4.6	-2.8	-7.0
Mean Daily Temperature Range	5% DB	MDBR	8.4	10.1	12.7	11.5	11.0	10.5	9.1	8.8	8.4	6.6	7.5	7.6
		MCDBR	7.3	7.6	11.6	12.6	14.9	14.3	13.5	13.7	12.5	6.7	5.9	7.8
		MCWBR	6.6	6.6	9.2	7.5	7.0	5.8	5.7	6.9	7.3	4.9	5.2	7.2
	5% WB	MCDBR	7.4	7.8	10.3	11.5	13.1	12.1	11.8	10.9	10.3	6.0	5.8	7.7
		MCWBR	6.7	6.9	8.2	7.1	6.3	5.3	5.3	5.7	6.2	4.6	5.2	7.1
Clear-Sky Solar Irradiance	taub	0.172	0.247	0.257	0.339	0.340	0.365	0.366	0.368	0.307	0.279	N/A	0.176	
	taud	1.937	2.206	2.283	2.122	2.406	2.371	2.410	2.417	2.558	2.396	N/A	1.931	
	Ebn at Noon	386	673	844	821	869	849	837	797	792	634	N/A	123	
	Edh at Noon	23	54	78	119	99	105	99	90	63	46	N/A	8	
All-Sky Solar Radiation	RadAvg	0.16	0.82	2.67	4.69	5.40	5.58	4.68	3.50	2.22	0.88	0.24	0.05	
	RadStd	0.02	0.09	0.20	0.32	0.58	0.47	0.34	0.36	0.27	0.12	0.03	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page